

USER MANUAL



PlcData

Version 0.0.1

BIT, INT, and FLOAT input-register waits for the Doosan Task Editor

DOOSAN DART USER COMMAND V2 MODULE

Zwart Industrial Innovations

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Package com.zii.plcdata | Target SDK 6

Document information

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Release 0.0.1: Initial public release with BIT, INT, and FLOAT input-register wait commands, translated property screens, optional timeout popup, and Task Editor summaries.

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1. Overview

PlcData adds three waiting commands to the Doosan Task Editor. Each command reads a PLC input register every 0.1 seconds and completes when the current value equals the configured target.

Task Editor label	Range	Purpose
Wait IR[B]	BIT 0-63	Wait for ON or OFF
Wait IR[I]	INT 0-23	Wait for a signed 32-bit value
Wait IR[F]	FLOAT 0-23	Wait for an exact decimal value

Safety notice: PlcData is not a safety component. Never use it instead of safety-rated I/O, a safety PLC, protective stops, or emergency-stop circuitry.

2. Installation

- 1 Download **com.zii.plcdata_0.0.1.dm**.
- 2 Remove the former **com.zwart.wri** package if installed, otherwise two PlcData groups can appear.
- 3 Open DART-Platform and switch to the authority level required for module installation.
- 4 Select **Install Module** on the Home screen and choose the downloaded DM file.
- 5 Open Task Editor and browse to **User Commands - PlcData**.

3. Command selection

 **Wait IR[B]** PlcData, 0, ON, 1 s, STP

 **Wait IR[I]** PlcData, 0, 0, 1 s, STP

 **Wait IR[F]** PlcData, 0, 0, 1 s, STP

The green hourglass is supplied by the Task Editor. The tested platform version uses the same icon colour for all three User Commands.

4. Wait for Input Register Bit

Wait until a selected input-register bit becomes ON or OFF.

Wait for Input Register Bit

Wait until the selected register bit reaches the desired state.

Register

0

Choose an integer from 0 through 63.

ON

OFF

☐

 No timeout (None)

☒

 Number of seconds

Timeout

1

seconds

Decimal values are allowed.

☒

 Show timeout popup

Field	Description
Register	Input-register bit number 0 through 63.
ON / OFF	Desired state. The command completes when the live bit equals this state.
No timeout	Wait indefinitely until the state matches.
Number of seconds	Use a positive timeout. Decimal seconds are accepted.
Show timeout popup	Show an alarm-style popup when the timeout expires.

```
wait_input_register_bit(register, value, timeout=None, timeout_message=None)
```

5. Wait for Input Register Int

Wait until a selected PLC input-register INT equals the configured signed integer value.

Wait for Input Register Int

Wait until the PLC register equals the configured integer value.

Address

0

Choose an integer from 0 through 23.

Desired integer value

0

Enter a 32-bit integer.

☐ No timeout (None)

☒ Number of seconds

Timeout

1

seconds

Decimal values are allowed.

☒ Show timeout popup

Field	Description
Address	Input-register INT address 0 through 23.
Desired integer	Signed 32-bit value from -2147483648 through 2147483647.
Timeout	None for an indefinite wait, or a positive number of seconds.
Timeout popup	Optional alarm-style operator notification.

```
wait_input_register_int(address, value, timeout=None, timeout_message=None)
```

6. Wait for Input Register Float

Wait until a selected PLC input-register FLOAT exactly equals the configured decimal value.

Wait for Input Register Float

Wait until the PLC register equals the configured float value.

Address

0

Choose an integer from 0 through 23.

Desired float value

0

Decimal values are allowed.

- ☐

No timeout (None)
- ☒

Number of seconds

Timeout

1

seconds

Decimal values are allowed.

- ☒

Show timeout popup

Field	Description
Address	Input-register FLOAT address 0 through 23.
Desired float	Decimal target value.
Timeout	None for an indefinite wait, or a positive number of seconds.
Timeout popup	Optional alarm-style operator notification.

```
wait_input_register_float(address, value, timeout=None, timeout_message=None)
```

Exact equality: FLOAT compares values using exact equality. Ensure that the PLC and robot use the same representable value.

7. Task Editor summary explained

The Task Editor inserts the module name before the summary supplied by PlcData. The module name is display information and is not sent to the PLC.

 **Wait IR[B]** PlcData, 0, ON, 1 s, STP

 **Wait IR[I]** PlcData, 0, 0, 1 s, STP

 **Wait IR[F]** PlcData, 0, 0, 1 s, STP

PlcData, 0, ON, 1 s, STP

Position	Example	Meaning
1	PlcData	Installed module name; not a PLC parameter.
2	0	BIT register or INT/FLOAT address.
3	ON	Desired BIT state, INT value, or FLOAT value.
4	1 s	Timeout in seconds. None means no timeout.
5	STP	Show Timeout Popup is enabled. It does not mean stop task.

INT and FLOAT examples

PlcData, 0, 42, 5 s, STP
PlcData, 0, 12.5, None

For INT and FLOAT, the second value is the address and the third value is the desired numeric value.

8. Wait-function reference

```
wait_input_register_bit(register, value, timeout=None, timeout_message=None)
wait_input_register_int(address, value, timeout=None, timeout_message=None)
wait_input_register_float(address, value, timeout=None, timeout_message=None)
```

Common execution flow

- 1 Read the configured input register.
- 2 Compare the current value with the configured target.
- 3 Return **True** immediately when the values match.
- 4 When they do not match, wait 0.1 seconds and read again.
- 5 If a positive timeout expires, optionally display the timeout popup and return **False**.

Return value: The generated Task Editor command calls the function as a statement, so its boolean result is not stored. Custom DRL code can call these functions directly and use True or False for branching.

Timeout behaviour

- None waits indefinitely until the target is reached or the task is stopped externally.
- Timeout values must be greater than zero; decimal seconds are supported.
- The final polling delay is shortened when less than 0.1 seconds remains.
- The popup is shown only when Show timeout popup is enabled.
- STP means Show Timeout Popup and is not a task-stop command.

Typical generated calls

```
wait_input_register_bit(0, ON, 1, message)
wait_input_register_int(0, 42, 5, message)
wait_input_register_float(0, 12.5, None, None)
```


9. PLC integration and safe use

- Confirm that the PLC and robot use the same input-register mapping.
- Confirm signed INT interpretation and the byte order used by the wider PLC integration.
- Use FLOAT only when exact equality is appropriate.
- Choose a timeout that accounts for PLC scan time, network updates, and process delay.
- Avoid an indefinite wait where a missing PLC update could block the task permanently.
- Validate each command in a safe test environment before production use.

Safety: These commands are convenience logic for normal process control. They are not certified or intended as functional-safety commands.

10. Troubleshooting

Symptom	Cause and action
Two PlcData groups	Both com.zwart.wri and com.zii.plcdata are installed. Remove the former package.
Same green hourglass	The Task Editor provides its generic icon. This does not affect operation.
Short command labels	The Task Editor shortens long User Command names. Version 0.0.1 uses ten-character labels.
Wait never completes	Check register mapping, target value, live PLC data, and exact FLOAT representation.
No timeout popup	Select a positive timeout and enable Show timeout popup. STP must appear in the summary.

Supported interface languages

Dutch, English, German, Polish, Spanish, Italian, and Korean.

11. Distribution and verification

Item	Value
Module file	com.zii.plcdata_0.0.1.dm
File size	166246 bytes
SHA-256	718EC2F569730264BE015C0E95445DFA F23292C61E29C61FE11744668FFC231D
Package	com.zii.plcdata
SDK	6

12. Support further development

PlcData is published by Zwart Industrial Innovations. Donations support further development, testing, documentation, and maintenance and are appreciated as a thank-you for making the modules available.



Scan the PayPal QR code to support future module development.

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